• **DRIVEABILITY—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED WITH DIAGNOSTIC TROUBLE CODES (DTCS) P0455, P0456, P0457, P0442, P1442, P1443, P1450 OR P1455—ROTUNDA VACUTEC 522 LEAK DETECTOR SMOKE MACHINE USAGE AND PROCEDURES** • **EMISSIONS—EVAPORATIVE EMISSION SYSTEM—ROTUNDA VACUTEC 522 LEAK DETECTOR SMOKE MACHINE USAGE AND PROCEDURES**	**Article No. 01-7-8**

FORD: 1997-2000 CONTOUR
1997-2001 CROWN VICTORIA, ESCORT, MUSTANG, TAURUS
2000-2001 FOCUS
1997-1998 F-250 HD
1997-1999 F-250 LD
1997-2001 ECONOLINE, EXPEDITION, F-150, RANGER, WINDSTAR
1997-2002 EXPLORER
1999-2001 SUPER DUTY F SERIES
2000-2001 EXCURSION
2001 ESCAPE

LINCOLN: 1997-1998 MARK VIII
1997-2001 CONTINENTAL, TOWN CAR
2000-2001 LS
1998-2001 NAVIGATOR

MERCURY: 1997-1999 TRACER
1997-2000 MYSTIQUE
1997-2001 GRAND MARQUIS, SABLE
1999-2001 COUGAR
1997-2002 MOUNTAINEER

ISSUE

Some vehicles may exhibit a Malfunction Indicator Lamp (MIL) illuminated with Diagnostic Trouble Codes (DTCs) P0455, P0456, P0457, P0442, P1442, P1443, P1450 or P1455 for the Evaporative Emission system. This may be caused by loose or leaking hoses, loose or broken fuel cap, etc... A leak at the top of the fuel filler pipe neck between the filler pipe and plastic insert is often misdiagnosed as a fuel filler cap leak.

ACTION

Verify condition is caused by the Evaporative Emission (EVAP) system. If necessary, use the Rotunda VACUTEC 522 Leak Detector Smoke Machine to help locate and diagnose leaks in the Evaporative Emission system. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Glossary Of Terms:

- **EVAP** - Evaporative Emission
- **VMV** - Vapor Management Valve
- **CVS** - Canister Vent Solenoid
- **NGS** - New Generation Star Tester Diagnostic Tool
- **WDS** - Worldwide Diagnostic System Diagnostic Tool

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Verify the concern is an Evaporative Emission System leak by performing the following:

1. Visual Inspection: Inspect the vehicle to determine if it has any of the following mechanical concerns.
   a. Improperly installed fuel filler cap.
   b. A leak at the top of the fuel filler pipe neck between the filler pipe and plastic insert is often misdiagnosed as a fuel filler cap leak.
   c. Loose or disconnected hoses - Example: VMV - vacuum hose to/from engine vacuum source port.
   d. Loose or disconnected components.
   e. Damaged EVAP service test port.
   f. Damaged EVAP canister or canister vent solenoid.
2. Reinstall or replace all evaporative emission hose clamps removed or damaged during the test procedures.
3. If the leak is NOT detected during the visual inspection, install the NGS or WDS Diagnostic tool to determine what DTC is stored in memory by using the appropriate test procedure by vehicle and model year in the Powertrain Control/Emissions Diagnosis (PC/ED) Workshop Manual.
4. Once it is verified by the NGS or WDS Diagnostic tool that one or more of the following DTCs are present: P0442, P0455, P0456, P0457, P1442, P1443, P1450 and P1455, perform the following test procedures.

NOTE
IF NONE OF THE EVAP LEAK DTCs MENTIONED IN STEP 4 ARE DETECTED, REFER TO THE PC/ED WORKSHOP MANUAL FOR THE APPROPRIATE VEHICLE AND MODEL YEAR DIAGNOSTICS.

ROTUNDA'S VACUTEC 522 SMOKE DETECTOR: Smoke Information

The Smoke Leak Detector injects a non-toxic DIAGNOSTIC VAPOR (Smoke) into the EVAP sealed vehicle system which is to be tested. Within seconds, smoke will begin to escape from the point or points of leakage. The following procedure will help determine where the leaks are located and help indicate what repairs are required.

NOTE
MANY EVAP LEAKS ARE CAUSED BY A LOOSE OR FAULTY FUEL CAP (P0457). LEAVE THE FUEL CAP IN PLACE WHEN COMPLETING THE SYSTEM LEAK CHECK PROCEDURE. IF THE VEHICLE FAILS THE SYSTEM SMOKE TEST PROCEDURE AT THE FUEL CAP, THE FUEL CAP CAN BE REPOSITIONED OR REPLACED. THE SYSTEM SMOKE TEST PROCEDURE WILL NEED TO BE REPEATED. THIS ACTION WILL ISOLATE THE FUEL CAP FROM THE REST OF THE EVAP SYSTEM AS A POTENTIAL CONCERN.

ROTUNDA'S VACUTEC 522 SMOKE DETECTOR: Power Supply Information

1. Connect the RED power cable to a 12-volt DC power supply. If you are using a battery, be sure it is in good condition and fully charged.
2. Connect BLACK GROUND CABLE to vehicle CHASSIS GROUND.

CAUTION
DO NOT CONNECT BLACK CABLE TO BATTERY GROUND. A SPARK IN THE VICINITY OF THE BATTERY MAY CAUSE AN EXPLOSION.

3. The "Power Indicator Lamp" should be on, indicating a good battery contact.

SMOKE LEAK DETECTOR MACHINE SERVICE PROCEDURE - ROTUNDA'S VACUTEC 522 (Model #218-00001)

Follow the test procedures listed below or in the Operator's Manual for the Vehicle Evaporative Emission Leak Detector System Smoke Machine - Rotunda's VACUTEC 522. This test procedure is located in the Operator's Manual under "Evaporative System Test and Diagnostics".

SYSTEM LEAK CHECK (METERED AIR SYSTEM) PHASE 1

1. Position the control lever (selector valve) to "METER" located on the VACUTEC panel.
2. Verify calibration of the flow meter by using one of the calibrated leak orifices (.051mm or 1.02mm - 0.020″ or 0.040 ″) - EVAP System Standard. Insert air supply hose (transparent hose) nozzle tip into appropriate EVAP System Standard located on the front of the VACUTEC panel.

3. TURN ON: Depress the remote starter button. "Observe" the position of the flow meter indicator ball.

4. Position the flow meter red "pointer flag" so it aligns with the measurement of the indicator ball observed in Step 3. Release the button and remove the air supply hose nozzle tip from the EVAP System Standard.

NOTE
THIS FLOW METER MEASUREMENT INDICATES THE VEHICLE PASS OR FAIL CRITERIA FOR THAT PARTICULAR LEAK ORIFICE EVAP SYSTEM STANDARD.

5. Vehicles with Non-VMV Quick Connector Hose: Remove VMV hose at engine vacuum source and plug hose.

6. Vehicles with VMV Quick Connector Hose: Remove VMV Quick Connector Hose at VMV port and cap VMV port.

7. Locate the vehicle's Evaporative Emission Service Test Port and remove the green cap. The EVAP Service Test Port is located close to or on the VMV.

8. Install EVAP Emission Test Port Adapter which is provided with the Rotunda's VACUTEC Smoke Machine (Part #218-00007).

9. Connect the transparent air supply hose nozzle tip into the EVAP Emission Service Test Port.

10. Use NGS or WDS (DataLogger) Diagnostic tool to CLOSE the Canister Vent Solenoid (CVS).

For WDS, refer to **How to Command the EVAP Canister Vent Solenoid Manually Using WDS DataLogger** later in this article and Figure 1.

For NGS Diagnostic Tool, refer to **Test Procedure for "Close the CVS"** later in this article.

NOTE
THE CVS CAN BE CLOSED FOR "9 MINUTES OR LESS".

11. Pressurize (metered air) the EVAP System: Depress the remote starter button on the Rotunda's VACUTEC Smoke Leak Detector. Observe if there is any flow (leak) on the flow meter gauge.

12. If NO flow is detected, the EVAP system is okay. The test procedure is complete. STOP. Remove the Smoke Machine Test equipment from the vehicle and hook up the VMV (Non-Quick Connector/Quick Connector) hose to or from its vacuum source.

13. If flow is detected, the EVAP system has an unacceptable leak. Proceed to SYSTEM SMOKE TEST PROCEDURE.

SYSTEM LEAK CHECK (SMOKE TEST PROCEDURE) PHASE 2

NOTE
A NON-TOXIC DIAGNOSTIC VAPOR (VISIBLE SMOKE) WILL BE INTRODUCED DURING THIS PROCEDURE.

NOTE
THE 2000/2001 LINCOLN LS HAS A UNIQUE EVAP SYSTEM. TO TEST THE 2000/2001 LINCOLN LS FUEL FILLER CAP SYSTEM FOR EVAP LEAKS, USE THE APPROPRIATE ROTUNDA THREADED FUEL CAP AND RECEIVER ASSEMBLY (PART #134-01039 - FORMERLY RELEASED AS PART# 134-00056) WHICH IS A COMPONENT SUPPLIED FROM ROTUNDA'S EVAPORATIVE EMISSION SYSTEM LEAK TESTER KIT.

1. Position the control lever (selector valve) on the Rotunda's VACUTEC 522 Control Panel to "SMOKE".

NOTE
LEAVE THE EVAP EMISSION PORT ADAPTER ON THE SERVICE TEST PORT FOR THE NEXT STEP.

NOTE
CHECK TO MAKE SURE THAT THE VMV (NON-QUICK CONNECTOR - REMOVE VMV HOSE AT ENGINE VACUUM SOURCE AND PLUG HOSE) OR (QUICK CONNECTOR - REMOVE VMV QUICK CONNECTOR HOSE AT VMV PORT AND CAP VMV PORT) IS IN THE PROPER POSITION.

2. Connect the black smoke hose nozzle tip to the EVAP Emission Test Port Adapter.

NOTE
THE CANISTER VENT SOLENOID NEEDS TO BE CLOSED BEFORE PERFORMING THE FOLLOWING STEPS. USE THE NGS OR WDS (DATALOGGER) DIAGNOSTIC TOOL TO CLOSE

CVS. REFERENCE THE FOLLOWING WDS OR NGS PROCEDURES TO CLOSE THE CVS. KEEP CVS CLOSED FOR "9 MINUTES OR LESS" DURING THIS TEST PROCEDURE.

3. Pressurize the EVAP Emission System with non-toxic diagnostic vapor (smoke). Depress remote starter button: Smoke will be introduced into the EVAP Emission System.
4. Hold the remote starter button down and use the halogen light provided with the VACUTEC 522 Leak Detector Smoke Machine to look for smoke in the engine compartment, EVAP hoses to VMV, CVS, carbon canister and area around the fuel cap. If a leak is detected, repair as necessary. Repeat EVAP LEAK SMOKE PROCEDURE Steps 2-4 to verify repair.
5. If the leak is not detected from the top, check under the bottom of vehicle. Inspect the EVAP hoses running to the carbon canister, fuel filler neck, CVS, and VMV. Repair the leak and repeat SYSTEM SMOKE TEST PROCEDURE Steps 2-5 to verify repaired leak concern.
6. Remove Smoke Machine equipment provided. Reinstall green cap to the EVAP Emission Test Port. Uncap and reinstall the VMV Non-Quick Connector or Quick Connector to engine vacuum source or VMV vacuum port. Recheck all electrical connections for the EVAP Emission System. Check all EVAP hoses for proper connection. This Smoke Machine Test Procedure is complete.

How to Command the EVAP Canister Vent Solenoid Manually Using WDS DataLogger:

1. Identify the vehicle with WDS.
2. Select Toolbox, then DataLogger.
3. Select Powertrain, then Engine.
4. From the DataLogger, PID selection (Figure 1, View A).
5. Select EVAPCV#.
6. Select units of "MODE" at the bottom of screen.
7. Select "Live Display" (Figure 1, View B).
8. Select EVAPCV# at the top of the screen, press "OSC Mode" (Figure 1, View C).
9. Press "Control Item Activate" (Figure 1, View D). EVAPCV# will turn to EVAPCV=. This means it is now under your control.
10. Select EVAPCV= and press "Control Item Increment" (Figure 1, View E) EVAPCV= will now turn "ON" (Closed). The EVAP system should now be sealed and can be tested for leaks. Selecting (Figure 1, View F) will turn EVAPCV= "OFF" (Open).

NOTE
THE CVS CAN BE CLOSED FOR "9 MINUTES OR LESS".

NGS Diagnostic Tool: Test Procedure for "Close the CVS"

1. Connect the scan tool.
2. Select vehicle.
3. Select "Diagnostic Data Link".
4. Select "PCM".
5. Select "Active Command" Modes.
6. Select the "Output Test" Mode.
7. Select the EVAPCV (EVAP Canister Vent) PID.
8. Select the "ALL OFF" Mode.
9. Close the canister vent solenoid by pushing the START button on the scan tool.

NOTE
THE CVS CAN BE CLOSED FOR "9 MINUTES OR LESS".

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 404000, 623000, 698298

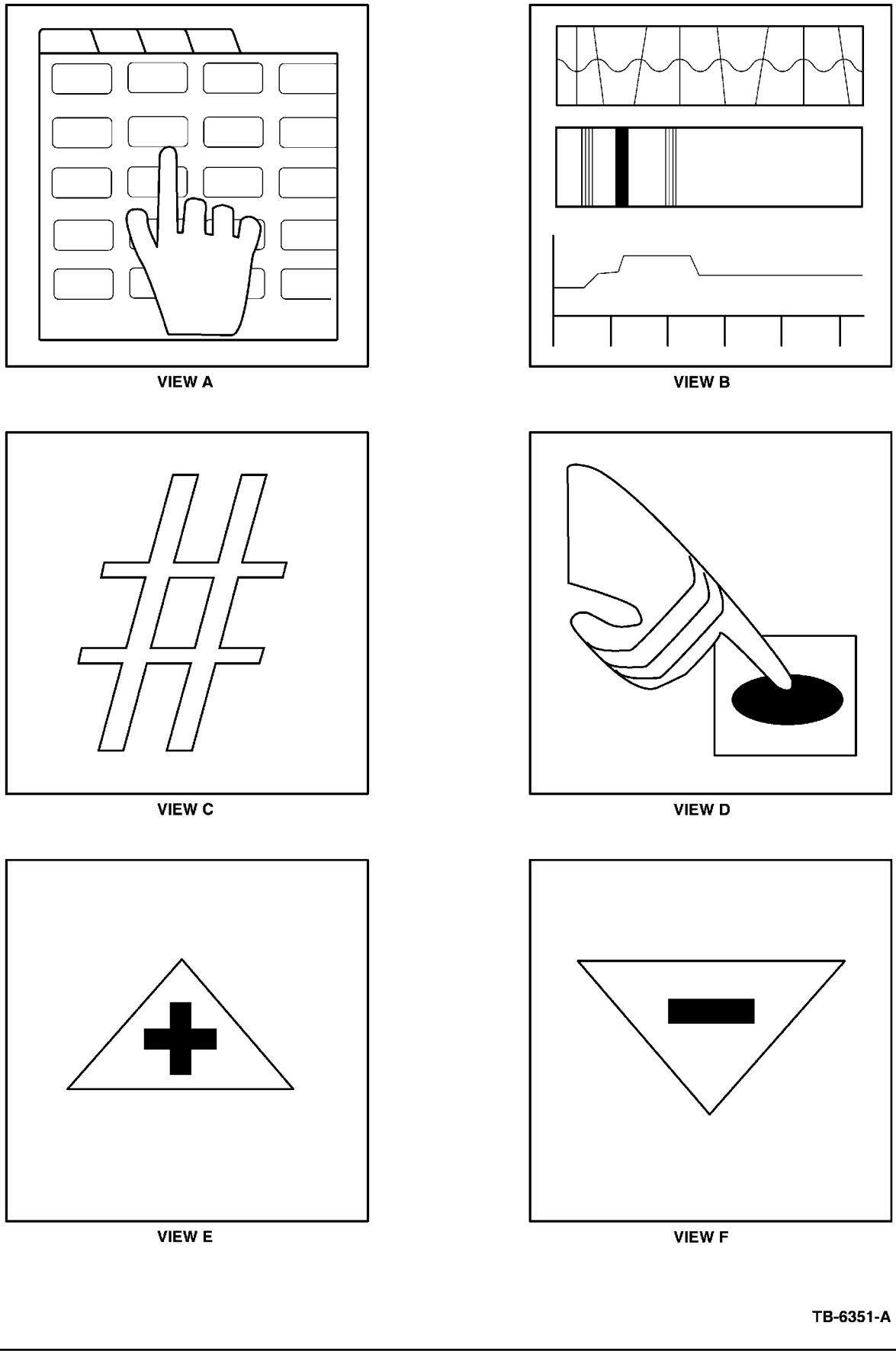

Figure 1 - Article 01-7-8